

Employment in First-Episode Psychosis: A Systematic Review and Bayesian Meta-Analysis

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REVIEW TITLE AND BASIC DETAILS

Review title ^{1 change}

Employment in First-Episode Psychosis: A Systematic Review and Bayesian Meta-Analysis

Review type

None

Condition or domain being studied ^{1 change}

Psychosis; Employment

Employment, educational participation, vocational functioning, and vocational recovery in first-episode

psychosis or early psychosis within five years of onset. The quantitative synthesis focuses on paid or

competitive employment prevalence and the effect of Individual Placement and Support on competitive

employment.

Rationale for the review

First Episode Psychosis (FEP) typically emerges during late adolescence or early adulthood, coinciding with a critical developmental period characterised by educational attainment, vocational identity formation, and transition into employment roles (Drake et al., 2013). The onset of psychosis during this stage can disrupt educational progression, occupational participation, and psychosocial functioning, often resulting in long-term vocational disadvantage and exclusion (Dudley et al., 2014). Accordingly, impaired vocational functioning has become increasingly recognised as one of the most enduring consequences associated with early psychosis. Despite many individuals with FEP expressing aspirations to engage in employment, education, or vocational training, unemployment and educational disengagement remain disproportionately elevated relative to the general population (Craig et al., 2014).

Emerging evidence indicates that vocational functioning may represent a clinically meaningful indicator of recovery trajectory. Employment and educational engagement have been associated with improved psychosocial adjustment and quality of life among individuals with

FEP (Drake et al., 2013). Conversely, vocational disengagement appears associated with poorer symptomatic and functional outcomes (Drake et al., 2013). Estimates indicate that employment rates among individuals with FEP range from approximately 13% to 55%, underscoring difficulties in achieving vocational recovery (Dewa et al., 2012). Consequently, vocational rehabilitation has become a prominent component of recovery-oriented Early Intervention in Psychosis (EIP) services. Individual Placement and Support (IPS), an evidence-based supported employment intervention, has demonstrated effectiveness in improving competitive employment outcomes among individuals with early psychosis (Bond et al., 2026). Therefore, a systematic review and meta-analysis will quantify employment rates and associated vocational outcomes in individuals with FEP.

Review objectives ^{1 change}

The review has three linked objectives. First, it will synthesise factors associated with employment and

vocational functioning, the quality and sustainability of vocational engagement, and interventions

intended to improve vocational recovery in first-episode psychosis. Second, it will estimate the study-

level point prevalence of paid or competitive employment at the assessment closest to 12 months

within a 6 to 24 month window under the study-specific care and vocational-intervention conditions of

eligible samples. Third, it will estimate the effect of Individual Placement and Support added to usual

care, compared with usual care alone, on competitive employment in randomised trials.

Narrative and

quantitative results will be interpreted with target-specific risk-of-bias and certainty assessments.

Keywords

Early psychosis; First-episode psychosis; Employment; Occupational engagement; Vocational outcomes

Country

England

ELIGIBILITY CRITERIA

Population ^{1 change}

Included

Inclusion: Participants with first-episode psychosis or early psychosis within five years of psychosis onset, diagnosed using a recognised diagnostic system or validated assessment, in inpatient, outpatient, community, or early-intervention settings. Reports containing broader or mixed populations will be eligible only when an extractable early-psychosis subgroup is available.

Excluded

Exclusion: Clinical high-risk states without a psychotic disorder, drug-induced or organic psychosis, bipolar disorder with psychotic features, and chronic schizophrenia without an eligible early-psychosis subgroup will be excluded.

Intervention(s) or exposure(s) 1 change**Included**

For the broad review, eligible exposures include employment status, educational or vocational engagement, occupational functioning, and vocational interventions. For the primary intervention meta-analysis, the intervention is Individual Placement and Support added to treatment as usual or usual early-intervention care.

Excluded

Trials of IPS plus an additional active component, comparisons with active vocational rehabilitation, non-vocational interventions, and non-randomised comparisons will be reported separately and will not enter the primary IPS-versus-usual-care synthesis.

Comparator(s) or control(s) 1 change**Included**

The primary intervention comparator is treatment as usual or usual early-intervention care without

Individual Placement and Support. Studies without comparators may contribute to the narrative review

and to the prevalence synthesis when they report an eligible employment proportion. Active vocational

comparators and within-cohort employed-versus-unemployed comparisons will be synthesised separately according to their estimands.

Study design 1 change

Both randomized and nonrandomized study types will be included.

Included

The broad systematic review includes peer-reviewed quantitative and qualitative primary studies. Reviews, meta-analyses, conference abstracts, book chapters, case reports, non-English reports, and unpublished or grey literature are excluded in accordance with version 1.0. The primary prevalence synthesis may include eligible cohorts and eligible study arms reporting a point prevalence. The primary intervention synthesis includes randomised controlled trials of IPS plus usual care versus usual care.

Other controlled and non-randomised designs will be tabulated or narratively synthesised but not pooled with the primary randomised comparison.

Excluded

- Studies including participants with psychotic disorders of more than 5 years in duration
- Drug induced psychosis or psychosis due to organic causes
- Studies that do not examine employment or vocational outcomes, or occupational functioning in individuals with FEP or early onset of psychosis
- Case studies

- Unpublished/grey literature
- Studies published in languages other than English will not be included.

Context

Participants with First-Episode Psychosis (FEP) or early psychosis from inpatient, outpatient, community, or Early Intervention in Psychosis (EIP) service settings.

TIMELINE OF THE REVIEW

Date of first submission to PROSPERO

25 May 2026

Review timeline

Review start date: 22 May 2026. Review end date: 31 December 2026.

Date of registration in PROSPERO

26 May 2026

AVAILABILITY OF FULL PROTOCOL

Availability of full protocol

A full protocol has been written and uploaded to PROSPERO. The protocol will be made available after the review is completed.

SEARCHING AND SCREENING

Search for unpublished studies

Only published studies will be sought.

Main sources that will be searched ^{1 change}

The main sources to be searched are *Embase*, *Google Scholar*, *MEDLINE*, *PsycINFO* and *PubMed*.

Other sources that will be searched

MEDLINE, Embase, Global Health, and APA PsycINFO were searched on the Ovid platform. PubMed and Google Scholar were searched directly. Searches covered database inception to 30 June 2026. Reference lists of included reports and relevant reviews were examined, and forward citation searching was undertaken.

Search language restrictions

The review will only include studies published in English.

Search date restrictions

Databases will be searched for articles published before 30 June 2026, there are no restrictions on search start date.

Other methods of identifying studies

Other studies will be identified by: *looking through all the articles that cite the papers included in the review ("snowballing" or forward citation searching).*

Link to search strategy

A full search strategy is available in the full protocol as described in the *Availability of full protocol* section

Selection process ^{1 change}

Studies will be screened independently by at least two people (or person/machine combination) with a process to resolve differences.

DATA COLLECTION PROCESS

Data extraction from published articles and reports

Data will be extracted by one person (or a machine) and checked by at least one other person.

Authors will not be contacted for further information.

Study risk of bias or quality assessment ^{1 change}

Risk of bias will be assessed using: *Cochrane RoB-2*

Data will be assessed by one person (or a machine) and checked by at least one other person.

Additional information will **not** be sought from study investigators if required information is unclear or unavailable in the study publications/reports.

Reporting bias assessment ^{1 change}

Missing-result bias will be assessed using ROB-ME or equivalent, separately for prevalence and intervention syntheses, with two reviewers and adjudication. Funnel/Egger/trim-and-fill from prevalence are limited to sensitivity due to proportion issues. For 3-trial synthesis, statistical assessment is unreliable; grey literature, trial registers, non-English reports are considered in judging missing-result risk.

Certainty assessment ^{1 change}

Certainty in the primary IPS effect will be assessed with GRADE, considering result-level risk of bias, inconsistency, indirectness, imprecision, and publication bias. Certainty in the pooled prevalence will be assessed with a transparent framework covering sampling frame and coverage, non-response and attrition, outcome measurement, risk of bias, indirectness, and between-setting inconsistency. Two reviewers will make and justify judgements, with adjudication. Certainty will be communicated wherever the corresponding results are interpreted.

OUTCOMES TO BE ANALYSED

Main outcomes ^{1 change}

The first primary quantitative outcome is study-level point prevalence of paid or competitive employment at the assessment closest to 12 months within a 6 to 24 month window, using the number assessed as the denominator. The pooled quantities are the median prevalence for an exchangeable study setting and, as a secondary descriptive summary, the model-implied mean prevalence across

exchangeable settings. These are not interpreted as person-weighted population prevalence.

The second primary quantitative outcome is competitive employment in randomised trials of

Individual Placement and Support plus usual care versus usual care, expressed as an available-case risk ratio. The exact arm-level binomial model reports an odds ratio as a sensitivity analysis. Absolute effects will be presented for pre-specified plausible control risks. The broad review also examines predictors of employment and vocational functioning, duration and quality of engagement, educational participation, associations with clinical and personal recovery, and intervention effects beyond the primary IPS comparison.

Additional outcomes ^{1 change}

Secondary and sensitivity analyses include period prevalence; a construct-relaxed point-prevalence set; reproduction of the original combined point and period prevalence set; exclusion of studies at high or weak appraisal ratings; exclusion according to verified vocational-intervention exposure; missing-as-not-employed bounds; normal-normal approximation to prevalence; leave-one-study-out analyses; alternative heterogeneity priors; a four-trial active-comparator package analysis; an exact arm-level trial model; and descriptive presentation of sparse recovery correlations, standardised mean differences, and non-comparable controlled comparisons. Exploratory meta-regressions examine design, assessment time, vocational exposure, and attrition on the original broader set, adjusted for outcome basis. These analyses will be identified as secondary or exploratory.

PLANNED DATA SYNTHESIS

Strategy for data synthesis ^{1 change}

The broad evidence will be synthesised narratively using the Synthesis Without Meta-analysis reporting

framework, organised around predictors of vocational outcome, quality and sustainability of engagement, and vocational interventions. Study design, outcome definition, follow-up, and risk of bias

will be considered when interpreting direction and magnitude.

The original frequentist analysis uses random-effects models in metafor. Employment proportions are

analysed on the logit scale with restricted maximum likelihood heterogeneity estimation and Knapp-

Hartung confidence intervals, with conventional Wald intervals reported where their small-sample

difference is informative. The primary intervention effect is a risk ratio for IPS plus usual care versus

usual care. Heterogeneity and prediction intervals are reported where estimable. These pooling methods were not specified in PROSPERO version 1.0 and are retained as the original analysis.

The amended Bayesian prevalence analysis uses a binomial-normal hierarchical model: employed

counts are binomial with a normal study effect on the logit scale. The pooled logit has a Normal(0, 1.5)

prior and the between-study standard deviation has a half-normal prior with scale 1, with scales 0.5 and 2 as sensitivities. The amended Bayesian trial analysis uses a normal-normal random-effects model on the log risk-ratio scale, with a Normal(0, 1) prior for the pooled log risk ratio and a half-normal prior with scale 0.5 for heterogeneity, with scales 0.25 and 1 as sensitivities. An exact arm-level binomial-logit model is secondary.

Results will emphasise effect magnitude, 95% credible intervals, heterogeneity, prediction intervals, posterior probabilities, absolute-effect scenarios, and sensitivity to priors and influential studies. MCMC diagnostics, prior predictive checks, posterior predictive checks, leave-one-study-out checks, and independent release reproduction will be required. A Dienes-style interval-null and directional half-normal Bayes-factor analysis will be reported as post hoc, exploratory, and prior-sensitive because the original results preceded its specification. The alternative scale and minimally interesting effect will be publicly justified, and robustness regions will be reported.

CURRENT REVIEW STAGE

Stage of the review at this submission 1 change

Review stage	Started	Completed
Pilot work	✓	✓
Formal searching/study identification	✓	✓
Screening search results against inclusion criteria	✓	✓
Data extraction or receipt of IPD	✓	
Risk of bias/quality assessment	✓	
Data synthesis	✓	

Review status

The review is currently planned or ongoing.

Publication of review results

Results of the review will be published in English.

REVIEW AFFILIATION, FUNDING AND PEER REVIEW

Review team members

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No conflict of interest declared.

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No conflict of interest declared.

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No conflict of interest declared.

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No conflict of interest declared.

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Funding source

Review has no specific/external funding but is supported by guarantor/review team (non-commercial) institutions.

Peer review

There has been no peer review of this planned review.

ADDITIONAL INFORMATION

Review conflict of interest

Declared individual interests are recorded under team member details.. No additional interests are recorded for this review.

Medical Subject Headings

Employment; Employment, Supported; Vocational Education; Psychotic Disorders

Revision note ^{1 change}

Major revision dated [2/09/26]. Version 1.0 specified a narrative synthesis and did not specify a statistical pooling model. After study selection and extraction, the review team conducted an original

frequentist random-effects meta-analysis of quantitative employment outcomes. This constituted a

deviation from the registered synthesis method. The original frequentist results were available before

the Bayesian analysis was specified. On 1 September 2026, the team approved a dated statistical

specification for a Bayesian reanalysis of employment prevalence and randomised Individual Placement

and Support trials, and no Bayesian numerical outputs were generated until after that approval.

Bayesian results were available before this major revision was prepared. The Bayesian analysis is

therefore pre-specified within the team and registered by amendment, but is not described as prospectively registered in PROSPERO. This revision adds the analysis objectives, estimands, likelihoods,

priors, sensitivity analyses, diagnostic checks, result-level risk-of-bias methods, and certainty assessments, and clarifies the search sources, review stage, and title. The original frequentist analysis

will be reported in full alongside the Bayesian analysis. All changes, their reasons, their dates, and

whether results were available are recorded in a dated amendment. The authors confirm no eligibility,

estimand, prior, threshold, missing-data rule, or sensitivity analysis was selected or changed because it

improved an observed result.

SIMILAR REVIEWS

Check for similar records already in PROSPERO

PROSPERO identified a number of existing PROSPERO records that were similar to this one (last check made on 22 May 2026). These are shown below along with the reasons given by that the review team for the reviews being different and/or proceeding.

- Supported education and IPS-like educational interventions for children and young people with mental health conditions: a systematic review (INSPIRE-Edu) [published 11 February 2026] [CRD420261305858]. The review was judged **not to be similar**
- Effectiveness of innovative interventions integrated into supported employment programmes for people with severe mental disorders: systematic review. [published 5 May 2026] [CRD420261338394]. The review was judged **not to be similar**
- Systematic review of interventions for employment participation in people with autism, intellectual disability, and/or psychosocial disability [published 9 December 2020] [CRD42020219192]. The review was judged **not to be similar**

PROSPERO version history 1 change

- [Version 2.0, published 05 Sep 2026](#)
- [Version 1.0, published 26 May 2026](#)

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